

# Financial engineering and derivatives



Financial derivatives have been around for centuries, but have grown in importance and sophistication in recent decades. Derivatives are contracts that involve an underlying commodity such as wheat or dollars – for example, a contract could specify a price and future date of delivery for 100 tonnes of barley. These kinds of contracts are particularly useful for managing risk: they can reduce uncertainty about future prices. But this feature of derivatives also allows speculators to make money. Suppose I buy a derivatives contract for the delivery of 100 tonnes of barley in six months for £100 per tonne. If the actual price of barley in six months is £120 tonnes per tonne I can make a profit of £20 per tonne by immediately selling on the barley. In recent decades, highly complex derivatives have been devised using mathematical models to price the risk inherent in different kinds of contracts. Some have blamed the recent economic crisis on

this kind of 'financial engineering', which introduced financial products so complex that few understood them.